

B.Sc. (I.T.) 6th Semester

Course: 602 ASP.NET Core MVC using C#

Course Code	602																								
Course Title	ASP.NET Core MVC using C#																								
Credit	4																								
Teaching per Week	4 Hrs																								
Minimum weeks per Semester	15 (Including Class work, examination, preparation, hold)																								
Last Review / Revision	June 2023																								
Purpose of Course	This course helps to learn basics programming development using ASP.NET Core and C#.NET Core.																								
Course Objective	The objective of the course is to impart basic introduction of C# language and concepts of web application development.																								
Course Out come	CO1: Students will be able to understand and learn C# using C#.NET Core. CO2: Students will be able to understand and develop web application development using ASP.NET Core MVC architecture using C#.NET Core. CO3: Students will be able to learn and implement data access using C#.NET Core for ASP.NET MVC Core based web application.																								
Mapping between COs with PSOs	<table><tr><td></td><td>PSO1</td><td>PSO2</td><td>PSO3</td><td>PSO4</td><td>PSO5</td></tr><tr><td>CO1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO3</td><td></td><td></td><td></td><td></td><td></td></tr></table>		PSO1	PSO2	PSO3	PSO4	PSO5	CO1						CO2						CO3					
	PSO1	PSO2	PSO3	PSO4	PSO5																				
CO1																									
CO2																									
CO3																									
Pre-requisite	Knowledge of Programming, Object Oriented Management System and Scripting languages like HTML, CSS, JavaScript, PHP, Python, etc.																								
Course Content	Unit : 1 : Introduction to .NET Core and C# 1.1. .NET Framework 1.1.1 .NET Framework architecture 1.1.2 Common Language Runtime 1.1.3 Common Type System 1.1.4 Common Language Specification 1.1.5 Microsoft Intermediate Language 1.1.6 Framework Class Libraries																								

Unit : 2 : Object Oriented Programming using C# .NET

- 2.1. Object Oriented Concepts and Principles
- 2.2. Classes
 - 2.1.1 Design of Class types
 - 2.1.2 Sealed Class
 - 2.1.3 Abstract Class and Abstract Base Class
 - 2.1.4 Partial Class
- 2.3. Structures
 - 2.2.1 Design of Structure type
 - 2.2.2 Comparison of Structure type and Class type
- 2.4. Enumeration
- 2.5. Construction and Disposal of object
- 2.6. Importance of special keyword like this, static
- 2.7. Access Modifiers
- 2.8. Inheritance
- 2.9. Polymorphism
 - 2.9.1 Overloading
 - 2.9.2 Overriding
- 2.10. Properties
 - 2.10.1 Importance of Property type
 - 2.10.2 Read-Only and Write-Only Properties
 - 2.10.3 Static Properties
- 2.11. Interfaces
 - 2.11.1 Concept and usage of Interface Type
 - 2.11.2 Designing Custom Interface
 - 2.11.3 Importance of Interface
 - 2.11.4 Comparing Interface with Abstract Base Class
 - 2.11.5 Implementing an Interface
 - 2.11.6 Working with built-in Interfaces like IEnumerable, ICloneable and IComparable
- 2.12. Exception & Error Handling
- 2.13. Threads and Multi-Threading
- 2.14. Nullable Type
- 2.15. Serialization
- 2.16. Attributes and Annotations

Unit : 3 : Working with Collections, Delegates and Events

- 3.1 Collections
 - 3.1.1 Introduction to Collections

- 3.3.3 Advantages of Delegates
- 3.3.4 Multicasting Delegates
- 3.3.5 Limitations of Multicast Delegates
- 3.4 Events
 - 3.4.1 Defining Events
 - 3.4.2 Creating Custom Events
 - 3.4.3 Publishing and Subscribing an Event
 - 3.4.4 Raising an Event

Unit : 4 : Programming using ASP.NET Core MVC

- 4.1. Introduction to ASP.NET Core
 - 4.1.1 Web Forms
 - 4.1.2 Web Pages
 - 4.1.3 MVC
- 4.2. Two-Tier, Three-Tier and N-Tier Application Architecture
- 4.3. MVC Architectural Pattern
- 4.4. URL Routing and Configuration
- 4.5. Working with Models
 - 4.5.1 Creating Models
 - 4.5.2 Data Access using Models
 - 4.5.3 Managing Temporary Data using Models
 - 4.5.4 Managing Data using ADO.NET
 - 4.5.5 Working with Stored Procedures
- 4.6. Working with Controllers
 - 4.6.1 Creating a Controller
 - 4.6.2 Role of Controller
- 4.7. Working with Views
 - 4.7.1 Creating and Rendering a simple View
 - 4.7.2 Custom View with and without Model
 - 4.7.3 Layout View and Partial View
 - 4.7.4 HTML Helper Class, Extension and Strongly Typed Methods
 - 4.7.5 Design of View using CSS
 - 4.7.6 Use of JavaScript and Bootstrap
 - 4.7.7 Working with Razor Pages
 - 4.7.8 Data Validation

Unit : 5 : Web Application Management

- 5.1 State Management
 - 5.1.1 Application
 - 5.1.2 Session State
 - 5.1.3 Cookie
 - 5.1.4 QueryString
 - 5.1.5 TempData, ViewBag and ViewData
- 5.2 Environment Variables
- 5.3 Multiple Environments and Development Mode
- 5.4 Unit Testing of MVC Application
- 5.5 NuGet Package Manager
- 5.6 Navigation